LABORATORY WORK 6

JOIN — PostgreSQL

Task 1

Question: Write a query that displays all flights of a specific airline.

Execution screenshot

Query | Query History | Scratc

```sql
SELECT
    f.flight_id,
    f.sch_departure_time,
    f.sch_arrival_time,
    a.airline_name
FROM flights AS f
JOIN airline AS a
    ON f.airline_id = a.airline_id
WHERE a.airline_name = 'Air Astana';
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No

	flight_id integer	sch_departure_time timestamp without time zone	sch_arrival_time timestamp without time zone	airline_name character varying (50)
1	1	2026-09-25 08:00:00	2026-09-25 10:00:00	Air Astana

Task 2

Question: Compose a query to obtain a list of all flights with the names of departure airports.

Execution screenshot

Query | Query History | Scratch P

```
1  SELECT
2      f.flight_id,
3      f.sch_departure_time,
4      dep.airport_name AS departure_airport
5  FROM flights AS f
6  JOIN airport AS dep
7      ON f.departing_airport_id = dep.airport_id
8  ORDER BY f.flight_id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 5 | Page No:

	flight_id integer	sch_departure_time timestamp without time zone	departure_airport character varying (50)
1	1	2026-09-25 08:00:00	Almaty International Airp...
2	2	2026-09-26 09:00:00	Astana International Airp...
3	3	2026-09-27 12:00:00	Istanbul Airport
4	4	2026-09-28 14:00:00	Dubai International Airport
5	5	2026-09-29 16:00:00	Frankfurt Airport

Task 3

Question: Create a query that finds all airlines that have no flights scheduled for the next month.

Execution screenshot

Query | Query History

```sql
SELECT
    a.airline_id,
    a.airline_name
FROM airline AS a
WHERE NOT EXISTS (
    SELECT 1
    FROM flights AS f
    WHERE f.airline_id = a.airline_id
      AND f.sch_departure_time >=
          DATE_TRUNC('month', CURRENT_DATE)
          + INTERVAL '1 month'
      AND f.sch_departure_time <
          DATE_TRUNC('month', CURRENT_DATE)
          + INTERVAL '2 months'
)
ORDER BY a.airline_name;
```

Data Output | Messages | Notifications

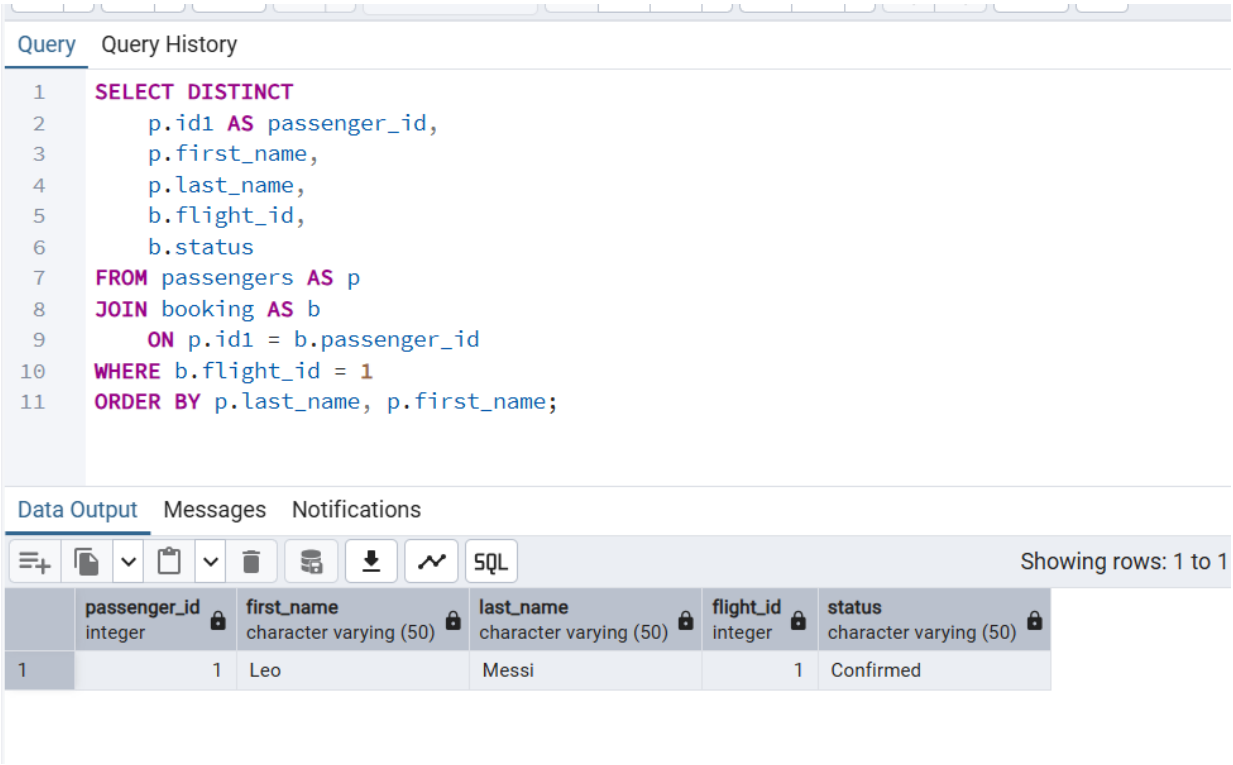
SQL

	airline_id [PK] integer	airline_name character varying (50)
1	1	Air Astana
2	7	AirEasy
3	8	Dana
4	4	Emirates
5	2	FlyArystan
6	9	FlyFly
7	5	Global Airways
8	6	KazAir
9	3	Turkish Airlines
10	10	wd

Task 4

Question: Create a query to display a list of passengers on a specific flight.

Execution screenshot



Query | Query History

```
1  SELECT DISTINCT
2      p.id1 AS passenger_id,
3      p.first_name,
4      p.last_name,
5      b.flight_id,
6      b.status
7  FROM passengers AS p
8  JOIN booking AS b
9      ON p.id1 = b.passenger_id
10 WHERE b.flight_id = 1
11 ORDER BY p.last_name, p.first_name;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1

	passenger_id integer	first_name character varying (50)	last_name character varying (50)	flight_id integer	status character varying (50)
1	1	Leo	Messi	1	Confirmed

Task 5

Question: Write a query that calculates the average, total, maximum and minimum price of tickets for each flight.

Execution screenshot

Query | Query History

```
1  SELECT
2      flight_id,
3      AVG(ticket_price) AS average_price,
4      SUM(ticket_price) AS total_price,
5      MAX(ticket_price) AS maximum_price,
6      MIN(ticket_price) AS minimum_price
7  FROM booking
8  GROUP BY flight_id
9  ORDER BY flight_id;
```

Data Output | Messages | Notifications

Showing ro

	flight_id integer	average_price numeric	total_price numeric	maximum_price numeric	minimum_price numeric
1	1	5086.2500000000000000	10172.50	10000.00	172.50
2	2	322.0000000000000000	322.00	322.00	322.00
3	3	517.5000000000000000	517.50	517.50	517.50
4	4	598.0000000000000000	598.00	598.00	598.00
5	5	218.5000000000000000	218.50	218.50	218.50

Task 6

Question: Create a query that shows all flights flying to a specific country by combining flights, airports and airline, and using the condition on the country name.

Query | Query History

```sql
SELECT
    f.flight_id,
    a.airline_name,
    dep.airport_name AS departure_airport,
    arr.airport_name AS arrival_airport,
    arr.country AS destination_country
FROM flights AS f
JOIN airline AS a
    ON a.airline_id = f.airline_id
JOIN airport AS dep
    ON dep.airport_id = f.departing_airport_id
JOIN airport AS arr
    ON arr.airport_id = f.arriving_airport_id
WHERE arr.country = 'Turkey'
ORDER BY f.flight_id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page N

	flight_id integer	airline_name character varying (50)	departure_airport character varying (50)	arrival_airport character varying (50)	destination_country character varying (50)
1	2	Turkish Airlines	Astana International Airp...	Istanbul Airport	Turkey

Task 7

Question: Display a list of minor passengers and their arrival destination.

Query | Query History

```sql
SELECT DISTINCT
    p.id1 AS passenger_id,
    p.first_name,
    p.last_name,
    dep.country AS departure_country,
    arr.airport_name AS arrival_airport,
    arr.country AS arrival_country
FROM passengers AS p
JOIN booking AS b
    ON p.id1 = b.passenger_id
JOIN flights AS f
    ON f.flight_id = b.flight_id
JOIN airport AS arr
    ON arr.airport_id = f.arriving_airport_id
JOIN airport AS dep
    ON dep.airport_id = f.departing_airport_id
WHERE p.date_of_birth > CURRENT_DATE - INTERVAL '18 years'
ORDER BY p.last_name, p.first_name;
```

Data Output | Messages | Notifications

passenger_id integer	first_name character varying (50)	last_name character varying (50)	departure_country character varying (50)	arrival_airport character varying (50)	arrival_country character varying (50)

Task 8

Question: Display the passenger's full name, passport number, and the passenger's current time of arrival at the destination.

Query | Query History

```
1  SELECT
2      p.first_name || ' ' || p.last_name AS full_name,
3      p.passport_number,
4      f.sch_arrival_time AS arrival_time
5  FROM passengers AS p
6  JOIN booking AS b
7      ON p.id1 = b.passenger_id
8  JOIN flights AS f
9      ON f.flight_id = b.flight_id
10 ORDER BY f.sch_arrival_time;
11
```

Data Output | Messages | Notifications

Showing

	full_name text	passport_number character varying (20)	arrival_time timestamp without time zone
1	Leo Messi	KZ100001	2026-09-25 10:00:00
2	Leo Messi	KZ100001	2026-09-25 10:00:00
3	Pele Marado...	KZ100002	2026-09-26 13:00:00
4	Yamal lamine	US100003	2026-09-27 15:00:00
5	Joan Garcia	ES100004	2026-09-28 18:00:00
6	Gerd Miller	DE100005	2026-09-29 18:00:00

Task 9

Question: Print a list of flights where the airline's home country and origin country are the same. Group them by the airport country.

Query | Query History

```sql
SELECT
    dep.country AS airport_country,
    f.flight_id,
    a.airline_name,
    a.airline_country
FROM flights AS f
JOIN airline AS a
    ON a.airline_id = f.airline_id
JOIN airport AS dep
    ON dep.airport_id = f.departing_airport_id
WHERE UPPER(a.airline_country) = UPPER(dep.country)
ORDER BY dep.country, f.flight_id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1

	airport_country character varying (50)	flight_id integer	airline_name character varying (50)	airline_country character varying (50)
1	Kazakhstan	1	Air Astana	KAZAKHSTAN

Task 10

Question: Write a query that lists each flight number, airline name, departure airport, and arrival airport by joining the relevant tables.

Query | Query History

```sql
SELECT
    f.flight_id,
    a.airline_name,
    dep.airport_name AS departure_airport,
    arr.airport_name AS arrival_airport
FROM flights AS f
JOIN airline AS a
    ON a.airline_id = f.airline_id
JOIN airport AS dep
    ON dep.airport_id = f.departing_airport_id
JOIN airport AS arr
    ON arr.airport_id = f.arriving_airport_id
ORDER BY f.flight_id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 5

	flight_id integer	airline_name character varying (50)	departure_airport character varying (50)	arrival_airport character varying (50)
1	1	Air Astana	Almaty International Airp...	Astana International Airp...
2	2	Turkish Airlines	Astana International Airp...	Istanbul Airport
3	3	Emirates	Istanbul Airport	Dubai International Airport
4	4	Global Airways	Dubai International Airport	Frankfurt Airport
5	5	FlyArystan	Frankfurt Airport	Almaty International Airp...

Task 11

Question: Display all passengers who have booked tickets for flights operated by a specific airline, including the passenger's full name and flight number.

Query | Query History

```sql
SELECT DISTINCT
    p.first_name || ' ' || p.last_name AS passenger_full_name,
    f.flight_id,
    a.airline_name
FROM passengers AS p
JOIN booking AS b
    ON p.id1 = b.passenger_id
JOIN flights AS f
    ON f.flight_id = b.flight_id
JOIN airline AS a
    ON a.airline_id = f.airline_id
WHERE a.airline_name = 'Air Astana'
ORDER BY passenger_full_name, f.flight_id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1

	passenger_full_name text	flight_id integer	airline_name character varying (50)
1	Leo Messi	1	Air Astana

Task 12

Question: Create a query that shows each flight's number, destination country, and number of passengers booked on that flight. Include flights with no booked passengers.

Query | Query History

```sql
SELECT
    f.flight_id,
    arr.country AS destination_country,
    COUNT(b.booking_id) AS passenger_count
FROM flights AS f
JOIN airport AS arr
    ON arr.airport_id = f.arriving_airport_id
LEFT JOIN booking AS b
    ON b.flight_id = f.flight_id
GROUP BY f.flight_id, arr.country
ORDER BY f.flight_id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 5

	flight_id integer	destination_country character varying (50)	passenger_count bigint
1	1	Kazakhstan	2
2	2	Turkey	1
3	3	United Arab Emirates	1
4	4	Germany	1
5	5	Kazakhstan	1

Task 13

Question: Find passengers who have not booked any tickets, using an appropriate JOIN operation.

Query | Query History

```sql
SELECT
    p.id1 AS passenger_id,
    p.first_name,
    p.last_name
FROM passengers AS p
LEFT JOIN booking AS b
    ON p.id1 = b.passenger_id
WHERE b.booking_id IS NULL
ORDER BY p.last_name, p.first_name;
```

Data Output | Messages | Notifications

	passenger_id integer	first_name character varying (50)	last_name character varying (50)

Task 14

Question: Display each airline and the total number of flights it operates, including airlines that currently have no flights.

Query | Query History

```sql
SELECT
    a.airline_id,
    a.airline_name,
    COUNT(f.flight_id) AS total_flights
FROM airline AS a
LEFT JOIN flights AS f
    ON f.airline_id = a.airline_id
GROUP BY a.airline_id, a.airline_name
ORDER BY a.airline_name;
```

Data Output | Messages | Notifications

	airline_id [PK] integer	airline_name character varying (50)	total_flights bigint
1	1	Air Astana	1
2	7	AirEasy	0
3	8	Dana	0
4	4	Emirates	1
5	2	FlyArystan	1
6	9	FlyFly	0
7	5	Global Airways	1
8	6	KazAir	0
9	3	Turkish Airlines	1

Task 15

Question: List flights departing from airports in a specified country and show the airline name, flight number, and departure airport city.

Query | Query History

```sql
SELECT
    a.airline_name,
    f.flight_id,
    dep.city AS departure_airport_city
FROM flights AS f
JOIN airline AS a
    ON a.airline_id = f.airline_id
JOIN airport AS dep
    ON dep.airport_id = f.departing_airport_id
WHERE dep.country = 'Kazakhstan'
ORDER BY f.flight_id;
```

Data Output | Messages | Notifications

Showing rows: 1

	airline_name character varying (50)	flight_id integer	departure_airport_city character varying (50)
1	Air Astana	1	Almaty
2	Turkish Airlines	2	Astana

Conclusion

This laboratory work practised SQL JOIN operations in PostgreSQL. INNER JOIN combined matching records across related tables, LEFT JOIN preserved rows without matching records, and aggregate functions with GROUP BY summarized ticket prices and passenger bookings. NOT EXISTS and IS NULL were used to identify airlines or passengers without related records.